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Project Title:	Mechanisms of Stochastic Resonance on Bone Cells		
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	Long-duration space flight induces bone loss. This loss could lead to a significant increase in risk of fracture. A specified form of mechanical stimulation (exercise), a combination of high-frequency vibration and low-frequency mechanical loading, may be an effective countermeasure to bone loss. This phenomenon is called stochastic resonance, in which noise vibration enhances the response of a nonlinear system to a weak signal. The global goal of this proposed study is to investigate mechanisms of stochastic resonance on bone cells. Understanding the mechanisms of stochastic resonance on bone cells will provide new information about increasing bone's response to mechanical stimulation. The hypothesis of the project is that stochastic resonance acts through L-type voltage sensitive calcium channels in the cell membrane to enhance bone response to mechanical loading.		

Task Description:	We will determine whether stochastic resonance increases mechanically sensitive genes and protein products through L-VSCCs in osteoblastic (MC3T3-E1) cells. We will study the effect of mechanical loading and stochastic resonance on mice that have an ablation of a specific calcium channel subunit (a subunit of L-VSCC) that is expressed in osteoblasts. We will study which mechanically sensitive genes are indirectly influenced by L-VSCCs during stochastic resonance exercise. Understanding the mechanisms of stochastic resonance on bone cells will provide new information about increasing bone's response to mechanical stimulation. The application of stochastic resonance offers potentially a new way to treat osteoporosis and prevent bone loss in space flight in the future. In addition, identification of the biochemical pathway(s) by which mechanical loading is transduced into gene expression may help to identify new drug targets for osteoporosis.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Space flight induces a suppression of bone formation in both cortical, cancellous bone and a reduction of bone mass. Periosteal bone growth decreases due to microgravity. Since bone mechanical properties are determined by both bone mass and geometry, loss of bone mass and reduced periosteal apposition resulting from space flight may lead to increased bone fragility and result in higher risk of fracture. Also, the changes in bone structure under microgravity mimic such clinical situations as skeletal unloading, prolonged bedrest, paralysis, cast immobilization. The related research can also benefit to increase our knowledge of the treatment of disuse or other osteoporosis.
	The research team has made progress to study effects of stochastic resonance and mechanical stimulation on bone cells. A. Effects of stochastic resonance on cultured bone cells In order to study the role of calcium channels in osteoblastic response to stochastic resonance, MC3T3-E1 osteoblastic cells were exposed to the following conditions: (1) Sine – an oscillating sinusoidal fluid flow at 2 Hz (a peak shear stress of 12 dynes/cm2), (2) Sine+Vibration – an oscillating sinusoidal fluid flow at 2 Hz of a decreased amplitude (a peak shear stress of 10 dynes/cm2) in combination with broad frequency vibrational components of 0-50 Hz (1.2 dynes/cm2 of the mean amplitude) to invoke stochastic resonance, or (3) Control – no exposure to fluid flow stimulation. The flow signals was generated using a home made oscillatory fluid flow system. Exposure to sine + vibration fluid shear elicited a significantly larger increase in activation of ERK1/2 and release of prostaglandins E2 (PGE2) over sine wave fluid shear alone. However, blocking the mechanosensitive cation-selective channels with administration of gadolinium inhibited the activational effect of vibration on ERK1/2 and PGE2 production. Inhibiting intracellular Ca2+ release by administering thapsigardin also significantly decreased the activation of ERK1/2 and PGE2 production induced by vibration. Furthermore, inhibition of the entry of calcium through the L-type voltage sensitive calcium channels with nifedipine significantly attenuated the osteoblastic response to both the sine wave and sine+vibration fluid shear stress. These data suggest that osteoblast sensitivity to stochastic resonance is mediated by Ca2+. B. Effects of stochastic resonance on cortical bone in vivo Mechanical loading of the skeleton is necessary to maintain bone structure and strength. Large amplitude strains associated with vigorous activity typically result in the greatest osteogenic response; however, data suggest that low-amplitude, broad-frequency vibration results in new bone formation and may enhance adaptation through a stochastic resonance (SR) phenomenon. That is, random noise may maximally enhance bone formation to a known osteogenic stimulus. The aims of this study were to (1) assess the ability of different vibration signals to enhance cortical bone formation during short- and long-term loading and (2) determine whether vibration could effect SR in bone. Two studies were completed wherein several osteogenic loading waveforms, with or without an additive low-amplitude, broad-frequency (0-50 Hz) vibration signal, were applied to the mouse ulna in axial compression. In study 1, mice were loaded short-term (30 s/day, 2 days) with either a carrier signal alone (1 or 2 N sine waveform), vibration signal alone [0.1 N or 0.3 N root mean square (RMS)] or combined carrier and vibration signal. In study 2, mice were loaded long-term (30 s/day, 3 days/week, 4 weeks) with a carrier signal alone (static or sine waveform), vibration signal alone (0.02 N, 0.04 N, 0.08 N or 0.25 N RMS) or combined carrier and vibration signal. Sequential calcein bone labels were administered at 2 and 4 days and at 4 and 29 days after the first day of loading in study 1 and 2, respectively; bone formation parameters and changes in geometry were measured. Combined application of the carrier and vibration signals in study 1 resulted in significantly greater bone formation than with either signal alone ($P < 0.001$); however, this increase was independently explained by increased strain levels associated with additive vibration. When load and strain levels were similar across loading groups in study 2, cortical bone formation and changes in geometry were not significantly altered by vibration. Vibration alone did not result in any new bone formation. Our data suggest that low-amplitude, broad-frequency vibration superimposed onto an osteogenic waveform or vibration alone does not enhance cortical bone adaptation at the frequencies, amplitudes and loading periods tested. C. Role of subunits, Cav1.2 and Cav1.3, of L-type calcium channels in load induced bone formation In order to investigate if L-type calcium channels mediate mechanically induced bone formation in vivo, mice with Cav1.3 knockout (KO) or tissue specific Cav1.2 KO mice were used. Cav1.2 and Cav1.3 are the subunits of L-type voltage sensitive calcium channels. C-1. Role of Cav1.3 subunit in bone formation (1) In the first study, age- and sex-matched Cav1.3 knockout (KO) and wild type (WT) mice are sorted into 3 groups of 8. Mice were subjected to in vivo axial compressive loading of the right ulna using a customized piezoelectric actuator. The animals will be loaded in 2Hz Sine wave with high (3 N), medium (2.5 N) and low (2 N) peak-to-peak amplitudes. The loading will be given to ulnae of mice for 60 seconds per day on 3 consecutive days. The left ulnae will serve as internal controls. During loading, mice will be anesthetized with isoflurane (0.5-2%) inhalation. Calcein (i.p. 7 mg/kg body mass) will be administered at day 5 and day 11 after the first load day. All animals will be sacrificed at day 18. After sacrifice, bone formation rates from undecalcified histological sections will be measured to assess the osteogenic potential of mechanical stimulation. Mean body weight of male Cav1.3 KO mice was lower than the male control. There was no significant difference in femur length among the groups. Femoral cross-sectional area and bone mineral content in male Cav1.3 KO mice decreased significantly in comparison with the male control. Loading induced new bone formation in both KO and WT animals compared with the left nonloaded ulnas. These data indicate that Cav1.3 knockout male mice have a lower body weight, smaller cross-sectional area, and less bone mineral content and density

Task Progress:

than wild type male mice. There was no significant difference between female knockout mice and wild type mice. Mechanically induced bone formation was not inhibited by knockout of cav1.3 in both male and female mice, suggesting Cav1.3 does not play a critical role in Mechanotransduction.

(2) The second study aimed to study if Cav1.3 plays a role in parathyroid hormone induced new bone formation. Parathyroid hormone (PTH) was given daily to both normal and Cav1.3 KO mice for 6 weeks. PTH increased bone mineral content in both control and KO animals. However, the increase of BMC in WT mice was higher than KO mice. Histological measurement is being carried out.

C-2. Role of Cav1.2 in bone formation

Conditioned Cav1.2 KO mice have been bred in our laboratory. Cav1.2 is the largest subunit of L-type voltage sensitive calcium channel. A tissue-selective knockout mouse is made by breeding the Cre mice with the floxed mice. Offspring that are homozygous for the Cre transgene and homozygous for the loxP sites will be deficient in Cav1.2 gene expression only in bone. This approach works because Cre recombinase cleaves the genomic DNA at the loxP sites and recombines the ends, effectively removing the DNA between the loxP sites. We have Col2.3-Cre and Col3.6-Cre mice in which Cre expression is driven by type I collagen promoter. Cre is not normally expressed in mammalian species so these mice will express Cre only in osteocytes and osteoblasts that also express type I collagen 2.3 fragment and 3.6 fragment, respectively. Therefore the expression of Cre will be limited to bone.

Analysis of bone phenotypes and mechanical loading experiment had been done. Deficiency of Cav1.2 in osteoblasts leads to lower body weights. Compared to the female control, the whole bone mineral density in female Cre2.3 driven Cav1.2 KO mice was lower. The whole bone mineral density in both male and female Cre3.6 driven mice was lower than the control. Histological measurements related to load induced bone formation using tissue specific Cav1.2 KO mice are being conducted at present.

D. Microarray analysis of the Molecular Profile after Mechanical Loading in Bone

Bone cells can sense physical forces and convert mechanical stimulation into biochemical signals that lead to gene expression. However, it is poorly understood how genes are systematically regulated in bone cells in response to mechanical loading. We have shown that one bout of axial loading of the ulna with a peak force of 20 N at 2 Hz for 360 cycles significantly increases periosteal bone formation in female SD rats compared with nonloaded control. Moreover, fatigue loading with a peak force of 20 N at 2 Hz until 10% stiffness loss generates substantial periosteal bone formation and microdamage in the cortex. In order to compare gene expression induced by loading and fatigue loading, we harvested ulna midshafts from control, loaded and fatigue loaded rats 6 hours following the loading regimens. RNA was extracted from those ulnas and used for gene expression assay using the Affymetrix U34 rat genome array. There were 4540 genes detected in the rat ulna. Those genes were subject to filtering at a significance level of $p<0.05$ and a fold change value of at least +/- 1.5 compared with the control. Both loading and fatigue loading resulted in a common set of 33 genes that were similarly regulated in magnitude and direction (26 genes increased and 7 decreased). Loading regulated an additional 38 unique genes (24 decreased and 14 increased) while fatigue loading specifically regulated another 122 genes (35 decreased and 87 increased). Canonical pathways analysis of the 3 sets of genes showed that bone anabolic and catabolic pathways were significantly enhanced in response to fatigue loading. Mechanical loading activated only anabolic pathways. The JAK/STAT (Janus kinase/signal transducer and activator of transcription) pathway was newly found to be involved in Mechanotransduction. Follow-up immunohistochenical staining of tissue sections revealed expression of STAT3 protein in the periosteal osteoblasts was greatly enhanced by mechanical stimulus. These results extend previous observations of gene expression changes and reveal in vivo regulation of genes specific to loading or fatigue loading.

E. A Systems Biology Informatics Approach to Compare Proteomics Profiles in Bone Induced by Loading and Fatigue Loading

Bone cells can sense physical forces and convert mechanical stimulation into biochemical signals that lead to expression of mechanically sensitive genes and proteins. However, it is poorly understood how genes and proteins are systematically regulated in bone cells in response to mechanical loading. In this study, we designed a systems biology approach to examine changes of global proteins and networks among the proteins by comparing Proteomics profiles among fatigue loading, loading and the control. We have shown that one bout of axial loading of the ulna with a peak force of 20 N at 2 Hz for 360 cycles significantly increases periosteal bone formation in female SD rats compared with nonloaded control. Moreover, fatigue loading with a peak force of 20 N at 2 Hz until 10% stiffness loss generates substantial periosteal bone formation and microdamage in the cortex. In order to compare Proteomics profiles induced by loading and fatigue loading, we harvested ulna midshafts and isolated proteins from the control, and loaded and fatigue loaded rats 4 days following the loading regimens. First, we used an experimental proteomics method based on a Label-free Liquid Chromatography / Mass Spectrometry (LC/MS) platform to obtain a list of 1,058 proteins from the total 5,361 proteins that are marked as “significantly changed” in any comparative analysis conditions. Homologous mapping of rat proteins to human proteins showed these 1,058 rat proteins matched 1,171 human proteins. We then focused on the proteins differentially expressed between loading and fatigue loading based on the fold change of threshold of 1.1 and further statistical analysis. A total of 42 proteins (from 21 Rat genes) were found significantly changed in fatigue loading group in comparison with the loading group. Next, we expanded these 42 proteins into a fatigue loading activated sub-network, which consisted of 394 human protein interacting pairs consisting of 297 human proteins. We examined sub-network protein interaction function categories using 2-dimensional visualization matrixes. We also developed a new visual representation structure for display of activated sub-networks using functional categories as network nodes and their crosstalk as network edges. We have found the activation of proteins and pathways relating to apoptosis and cell division in response to fatigue loading.

F. Low-Intensity Pulsed Ultrasound and a Nonsteroidal Anti-Inflammatory Drug Have Opposing Effects During Stress Fracture Repair

Low-intensity pulsed ultrasound (LIPUS) and non-steroidal anti-inflammatory drugs (NSAIDs) were used to treat stress fracture. Bilateral stress fractures were induced in the ulnas of 48 adult rats. Animals were divided into two groups (NSAID and VEH), and treated 5 days per week with celecoxib (5 mg/kg) mixed in a vehicle solution of polyethylene glycol and saline (NSAID) or vehicle alone (VEH). One-to-three hours following drug administration, all animals were treated with unilateral active-LIPUS and contralateral inactive-LIPUS. Equal numbers of ulnas from each drug group were histologically evaluated at two, four and eight weeks following induction of stress fracture. Neither LIPUS nor NSAID influenced bone resorption, but each had significant and opposite effects on intracortical bone formation rate.

	These effects indicate that LIPUS may be used to facilitate stress fracture repair whereas NSAID may delay tissue level repair of stress fractures. There was no interaction between LIPUS and NSAID, indicating that the beneficial LIPUS effect was not mediated by the cyclooxygenase-2 pathway. LIPUS accelerated stress fracture healing, whereas the NSAID delayed repair. When used in combination, the beneficial LIPUS effect was not impaired by the detrimental NSAID effect.
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